

SIMATS ENGINEERING

ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics

COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Manipulation (Level 2)

Total Marks: 25

Question Count: 5

Code of Conduct

I **LEO LEENAS A (192511068)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **LEO LEENAS A (192511068)**

Assessment Tasks

Scenario-Based Questions

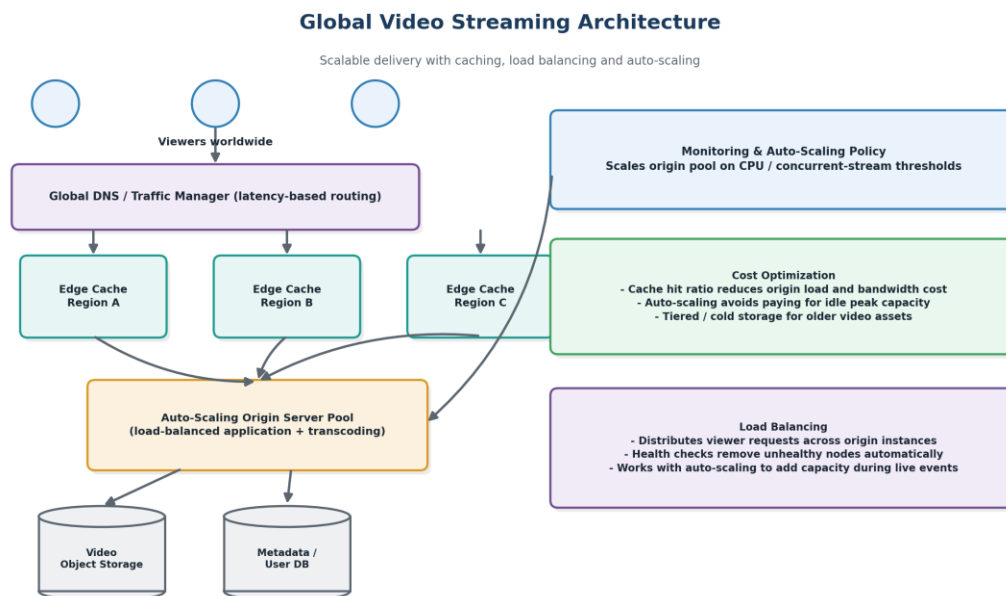
1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.
2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.
3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy. Design a solution using cloud-native backup, replication, and versioning. Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved. Provide steps to ensure business continuity during failures.
4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.
5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

Question 1: Cloud Architecture for a Global Video Streaming Platform

Scenario — A media company is migrating its streaming platform to the cloud and must support millions of concurrent viewers with minimal buffering.

The design places CDN edge caches in each major region so viewers pull video from a nearby location instead of a single origin server, which is the single biggest factor in cutting buffering. A global traffic manager routes each viewer to the nearest healthy edge node, and behind the edge layer sits an auto-scaling pool of origin and transcoding servers, kept behind a load balancer so no one instance gets overloaded and unhealthy instances are automatically taken out of rotation.

Auto-scaling adds origin capacity automatically during a big live event and shrinks it back once viewership drops, which is also where the cost savings come from: the company is not paying for peak-day capacity all year round. Combined with a high cache-hit ratio at the edge, which reduces how often the origin has to serve a video directly, and cheaper storage tiers for older content that is watched less often, the overall cost stays proportional to actual demand rather than worst-case demand.



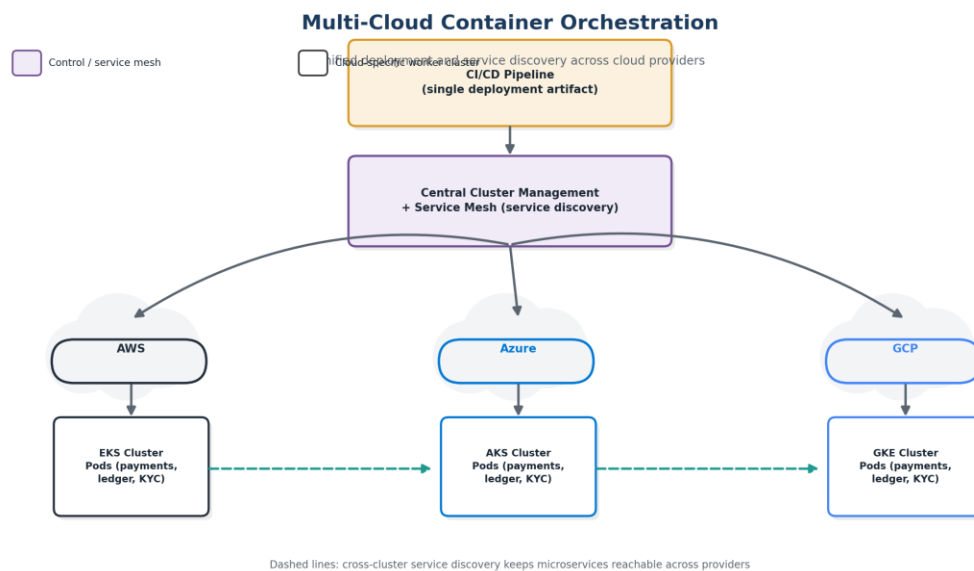
Global video streaming architecture with edge caching, load-balanced auto-scaling and cost controls

Question 2: Multi-Cloud Container Deployment for a Fintech Startup

Scenario — A fintech startup runs containerized applications across multiple cloud providers and is struggling to keep deployments consistent.

Kubernetes gives the team one deployment model that behaves the same way whether it runs on AWS, Azure, or GCP, so the same container images and manifests can be applied to any provider instead of writing separate deployment logic for each. A central management layer with a service mesh sits above the individual clusters and handles service discovery, so a payments service on one cloud can still find and call a KYC service running on another cloud through a stable internal name rather than a hard-coded address. Scaling is handled per cluster through the standard Kubernetes horizontal pod autoscaler, driven by CPU or request-rate metrics.

The main benefit of this multi-cloud approach is avoiding vendor lock-in and being able to place workloads closer to specific regulators or customers. The risk side is real too — running three different Kubernetes services means more operational complexity, inconsistent networking rules between providers, and higher management overhead, so this approach only pays off once the team has the tooling in place to manage it centrally rather than cloud by cloud.



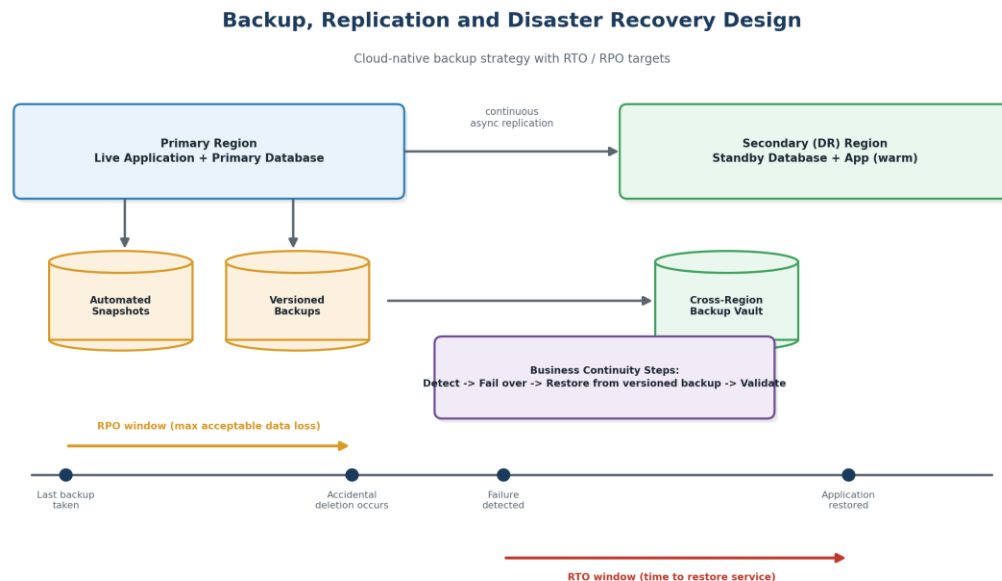
Multi-cloud container orchestration with a shared control plane and cross-cluster service discovery

Question 3: Backup and Disaster Recovery for an E-Learning Platform

Scenario — An e-learning platform lost data through accidental deletion and needs a proper backup and recovery strategy.

The fix combines three layers: automated snapshots taken on a fixed schedule, versioning turned on so an older copy of a file can be restored even after it has been overwritten or deleted, and continuous replication of the primary database to a secondary region so a full copy of the data exists somewhere else at all times. Recovery point objective is set by how often snapshots and replication run — if data is replicated every few minutes, the RPO is a few minutes of possible data loss at worst. Recovery time objective is set by how fast the standby environment in the secondary region can be brought online, which is much faster if that environment is kept warm rather than built from scratch after a failure.

Business continuity during a real incident follows a short, repeatable sequence: detect the problem through monitoring, fail over to the secondary region or restore the specific deleted item from a version, validate that the data is correct, and only then route users back to normal service. Testing this sequence periodically, not just writing it down, is what actually keeps the RTO and RPO numbers realistic.



Backup, cross-region replication and the RPO/RTO timeline around a failure event

Question 4: Hybrid Cloud Architecture for a Government Agency

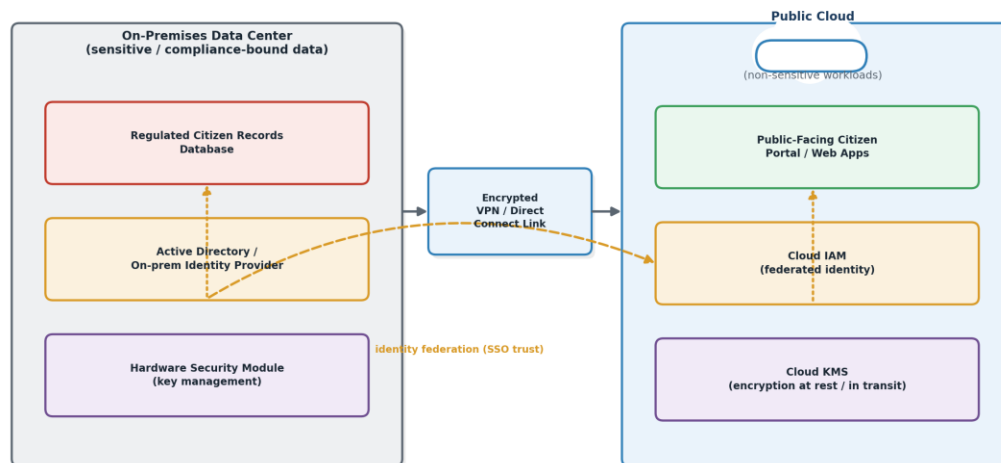
Scenario — A government agency must keep sensitive data on-premises for compliance while moving other workloads to the cloud.

Sensitive citizen records and the key management system stay inside the on-premises data center, since that is what the compliance requirement actually demands, while public-facing portals and non-sensitive workloads move to the cloud where they can scale more easily. The two environments are joined by an encrypted VPN or a dedicated private link rather than the open internet, so traffic between them is never exposed. Identity management is federated — the on-premises Active Directory stays the single source of truth for who a user is, and it is trusted by the cloud's identity service through single sign-on, so staff do not need separate accounts for each environment. Encryption is applied at rest and in transit on both sides, with the on-premises hardware security module and the cloud key management service each handling keys for their own data.

The hardest part of running this in practice is not the initial design but keeping it consistent over time — access policies, patching schedules, and audit logs have to be managed across two very different environments, and it is easy for one side to drift out of sync with the other unless both are governed by the same security policy from day one.

Hybrid Cloud Architecture for Sensitive and Non-Sensitive Data

Secure connectivity between on-premises and public cloud



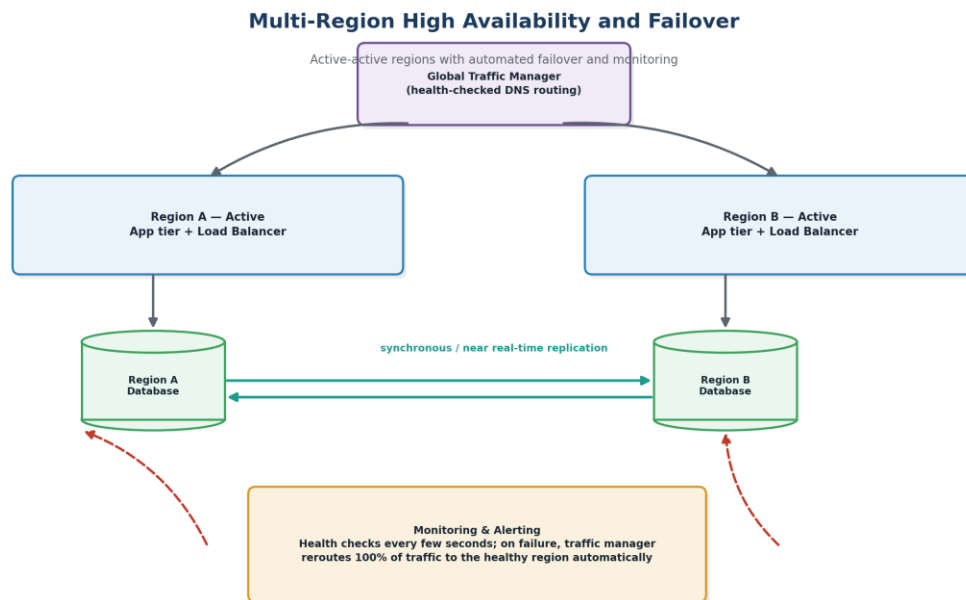
Hybrid architecture separating regulated on-premises data from cloud workloads through a secure link and federated identity

Question 5: Multi-Region High Availability for a SaaS Application

Scenario — A SaaS provider must keep its application running even if an entire cloud region fails.

Running the application active-active in two regions, each with its own load balancer and app tier, means neither region is sitting idle waiting for a disaster — both are already serving live traffic. A global traffic manager in front of both regions constantly health-checks them, and a database replication link keeps both regional databases in near real-time sync so that either region has a current copy of the data at all times. If Region A goes down, the traffic manager simply stops sending traffic there and Region B absorbs the full load, without anyone needing to manually trigger a failover.

Making that failover actually work depends on monitoring that checks more than just whether a server responds — request latency, error rates, and database replication lag all need their own alerts, because a region can be technically up while quietly falling behind on data. Catching that early, before it turns into a real outage, is what proactive monitoring is really for.



Active-active multi-region deployment with database replication, health checks and automatic failover

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

Total Marks Awarded: